

AGENTTRADE
A BULL-BEAR DEBATE DRIVEN MULTI-AGENT FRAMEWORK FOR
STOCK ANALYSIS AND RISK-CONTROLLED PAPER TRADING

PROJECT REPORT FOR THE COURSE
AD23731 FOUNDATIONS OF AGENTIC AI

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BONAFIDE CERTIFICATE

Certified that this is a Bonafide Record of the project work done by **Doneeswaran J. (231501502)** and **Nisha S. (231501112)** of VII Semester, B.Tech. Artificial Intelligence and Machine Learning Department in partial fulfilment of the requirements of the course **AD23731 – Foundations of Agentic AI** during the Academic Year 2026 – 2027.

Faculty in charge

Submitted for the Project Review held on:

INTERNAL EXAMINER

EXTERNAL EXAMINER

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1. INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Stock research brings together price trends, news, portfolio constraints and uncertainty. A conversational model can give a confident answer without showing the evidence behind it. For a college project, the important question is therefore how retrieval, structured reasoning and human review affect the answer, rather than how polished the chat interface looks.

AGENTTRADE makes those stages visible in a local research pipeline. It is intended for decision-support experiments, not autonomous investment management. US-listed stocks are used because the order integration is an Alpaca paper account. Paper trading simulates broker activity and must remain separate from real-money credentials [6].

1.2 NEED FOR AGENTIC AI

Retrieval-Augmented Generation (RAG) adds external evidence to model context [1]. Agentic reasoning introduces a choice of tools and queries instead of relying on one fixed context [2]. In this project, separate roles challenge an initial interpretation and a bounded correction pass revises it.

These features make the workflow easier to inspect, but they do not guarantee factual accuracy. Traces and source-support checks are still needed. Agentic AI is therefore used here as a design choice to be tested, not as proof of higher accuracy.

1.3 PROJECT OBJECTIVES

The primary objective is to develop a local, bounded multi-agent research assistant for US-listed stocks that separates evidence-based research from human-approved paper trading, and to evaluate how the retrieval policy affects its answers. The specific objectives are:

- To support named-stock research and open-ended discovery with readable explanations.
- To compare no retrieval, fixed retrieval and agent-selected retrieval on a frozen classroom task set.
- To compare Qwen2.5 3B and Llama3.2 3B under comparable single-model benchmark conditions.
- To separate model reasoning from deterministic proposal limits and explicit human approval.
- To provide the source, interface evidence, workflow diagrams and an honest account of results and limitations.

2. PROBLEM STATEMENT AND BUSINESS CASE

2.1 PROBLEM DEFINITION

An unstructured financial question can lead to a weakly supported recommendation, unfamiliar language or an accidental jump from analysis to execution. The user needs to see what was researched, which data was available, which limits apply and whether an order has actually been sent. The system must treat research, a pending proposal and a broker receipt as different states. The problem addressed by this project is therefore stated as follows:

“How can a stock-research question be answered with visible evidence, challenge and critique, while keeping research, pending proposals and paper-broker execution as separate, human-controlled states?”

2.2 LIMITATIONS OF EXISTING APPROACHES

The following table compares a single conversational model with the proposed system. The AGENTTRADE column describes design intent and does not claim improved investment accuracy.

Table 2.1: Comparison of stock-research approaches

Aspect	Single conversational model	AGENTTRADE
Evidence	May answer confidently without showing the evidence behind the answer	Dated source cards and computed indicators retrieved from Qdrant
Challenge	Internal challenge is hard to inspect	Separate bull, bear, critic and one bounded correction role, visible in execution logs
Risk limits	Not enforced by the model	Deterministic Python checks: up to 5 shares, USD 1,000 notional, no short selling, 30-minute expiry
Execution	Analysis can blur into action	Research has no order path; only explicit human approval reaches the paper-order path
Error handling	Failures may be hidden behind fluent text	Errors and unconfigured states are shown directly

2.3 BUSINESS AND REAL-WORLD CONTEXT

AGENTTRADE can be used as a decision-support system for stock research and risk-controlled paper trading. It helps users analyze market information, compare bullish and bearish perspectives, and review risks before making a trading decision.

The system can support retail investors, financial researchers, and trading platforms by providing structured and evidence-based analysis. For real-world deployment, it would require stronger security, licensed financial data, operational controls, and financial/legal compliance. The current system is intended as a research prototype and is not an investment adviser.

The following table maps each problem to the corresponding capability of AGENTTRADE.

Table 2.2: Problems and the proposed solution

Problem	AGENTTRADE solution
Weakly supported recommendation	Dated source cards with fixed or agent-selected retrieval
Unfamiliar financial language	Readable summary separated from technical detail
Accidental jump from analysis to execution	Separate proposal and approval endpoints; research has no order path
Research, proposal and receipt confused	Distinct states for research, pending proposal, broker receipt and broker-owned portfolio
Data outage disguised as a missing stock	Errors and unconfigured states displayed directly

3. USER AND STAKEHOLDER ANALYSIS

3.1 TARGET USERS

The primary users are the student project team and academic evaluators. A paper-account user can inspect research and choose whether to approve a simulated order. Future research users may use the evidence view, but must not read a model verdict as a guaranteed market outcome.

3.2 STAKEHOLDER IDENTIFICATION

Table 3.1: Stakeholders and their key needs

Stakeholder	Key needs
Retail Investor	Research stocks and evaluate recommendations
Financial Researcher	Analyze market data and supporting evidence
System Administrator	Manage models, data sources, and system configuration
Paper-Trading User	Review and approve paper-trading proposals

3.3 USER REQUIREMENTS

Functional requirements:

- Accept a natural question in a single research box; detect common company names and explicit tickers.
- Ask for a ticker when the input is unresolved instead of silently choosing a default, and ask for clarification when multiple distinct stocks are named.
- Start a named-stock request without an extra confirmation step.
- For open-ended discovery, present one candidate and up to two alternatives, with a warning that only a small screen was considered.
- Separate a readable summary from technical detail, and keep portfolio information, pending proposals and approval controls distinguishable.

Non-functional requirements:

- Transparency — errors and unconfigured states must be shown directly. A data outage must not be disguised as a missing-stock question, and absent paper credentials must not create an apparently executable proposal.
- Security hygiene — local services should stay bound to the host, and secrets should not appear in reports or source control.
- Traceability — execution traces should be retained for evaluation.

4. PROPOSED AGENTIC AI SOLUTION

4.1 OVERALL SOLUTION

The service combines a bounded six-role workflow with deterministic validation. Live evidence is prepared for valid supported US tickers. A separate synthetic AAPL/MSFT dataset is used for controlled classroom experiments. The two sources serve different purposes and should not be confused.

4.2 KEY FEATURES

Named-stock research. A question such as “Is Tesla a good buy?” resolves TSLA, validates the market data and starts research. A narrow alias map supports common names; unfamiliar companies may need an explicit ticker. The response is research text, not an order authorization.

Open-ended discovery. A question such as “What is the best stock to buy now?” fetches candidates from Yahoo most-active and top-gainer screens. The default considers up to three candidates, not the whole market. Each candidate is researched; a technical score breaks verdict ties. The response contains the closest candidate and up to two backups. The chosen candidate is not necessarily a BUY, so a guaranteed “best stock” claim is not supported.

In the current browser, an agentic retrieval selection is mapped to fixed retrieval for discovery. Named-stock agentic research remains available. Discovery coverage and the frozen benchmark must therefore be evaluated separately.

4.3 AGENT-BASED APPROACH

The retrieval, analyst, bull, bear, critic judge and correction roles generate evidence interpretations and judgments. Python checks market data and proposal limits. Proposal creation and approval are separate user actions and endpoints. Research cannot authorize an order, and only explicit approval can reach the paper-order path.

The interface includes dated source cards, computed indicators, summary and detail regions, proposal review and a broker-owned portfolio. The login interface is not production access control. The deployment remains a localhost educational prototype.

5. DATASET AND KNOWLEDGE SOURCES

AGENTTRADE combines a frozen synthetic classroom corpus with live public market data and a paper broker account. Each source is used for a specific purpose and has stated limits.

Table 5.1: Knowledge sources and limits

Source	Use and limitation
Frozen synthetic cards	AAPL/MSFT classroom facts and policies with IDs and dates. Not real financial statements.
25 frozen questions	Identity, policy, indicator and evaluation questions with expected label terms. A narrow benchmark.
Yahoo Finance	History, statistics, listing checks, discovery candidates and headline metadata; may be delayed or unavailable.
Optional NewsAPI	Headline source when a local development key is configured. Wider use requires a coverage and plan check.
Alpaca paper account	Account, holdings, buying power, orders and portfolio history. Required for paper execution.

Preparation and indicators. Python computes RSI14, SMA20/50, MACD and its signal line from price history. Cards retain source and date metadata. Nomic embeddings represent the text in Qdrant, and ticker tags allow company-scoped retrieval. A headline is not a full article and does not verify the underlying claim.

Data quality and reproducibility. Missing news produces a flagged price/statistics fallback. Synthetic policy card 3 contains an obsolete two-ticker restriction and is annotated as historical; current API rules define the live scope. The frozen benchmark should run on a clean Qdrant collection before live research, because ingest does not remove every previously added live card and mixed data may change the comparison. A formal experiment should preserve model versions, prompts, embeddings, the corpus and retrieval traces.

6. TOOLS AND TECHNOLOGIES USED

The technology stack combines a containerised API and vector store with natively run orchestration and inference services, so that the whole prototype can be inspected on one machine.

Table 6.1: Technology stack

Layer	Technology
Interface	HTML, CSS and JavaScript; research desk, shortcuts, Markdown output and TradingView widgets
API	FastAPI 0.116.1, Uvicorn 0.35.0, Pydantic 2.11.7 and HTTPX 0.28.1
Market calculations	yfinance 0.2.65 and pandas 2.3.2 for public data and indicators
Orchestration	Native n8n with separate research and approval workflow JSON
Inference	Native Ollama; qwen2.5:7b and :3b, llama3.2:3b for benchmarking, nomic-embed-text
Retrieval and state	Qdrant container and SQLite in a persistent ledger volume
Packaging	Docker Compose for API and Qdrant; host.docker.internal reaches native services
Broker	Alpaca paper API only; no real-trading endpoint

Deployment responsibilities. Docker builds the API from pinned requirements. Compose publishes ports 8000 and 6333 on 127.0.0.1 only. Native n8n typically uses port 5678 and Ollama uses 11434. Mac and Windows require the same models, with host-specific native setup. n8n provides orchestration [3], Ollama serves local inference [4], and Qdrant provides vector retrieval [5].

Practical constraints. Local models require RAM, storage and inference time. Chart and Markdown assets load from CDNs, so the interface is not fully offline. Market data and broker calls also need the internet. The local services should not be exposed publicly; the current secret and identity design is not suitable for that use.

7. AGENT ARCHITECTURE

7.1 AGENT COMPONENTS

Table 7.1: Agents, models and responsibilities

Role	Default model	Responsibility
Retrieval	Qwen2.5 3B	Choose an evidence search and inspect metadata
Analyst	Qwen2.5 7B	Interpret evidence and indicators; prepare an initial view
Bull case	Qwen2.5 3B	Develop bounded supporting arguments
Bear case	Qwen2.5 3B	Challenge uncertainty and weak assumptions
Critic judge	Qwen2.5 7B	Assess support, conflicting cases and the decision
Correction	Qwen2.5 3B	Revise once using critic feedback

Figure 7.1 separates the browser, API, native inference workflow, evidence store, local ledger and external paper broker. Model output is returned to the interface through the API. The paper-order route belongs to the separate human-approved path, not to the research agents.

AGENTTRADE | System architecture

Research proposes. Human approval is the only route to a paper order.

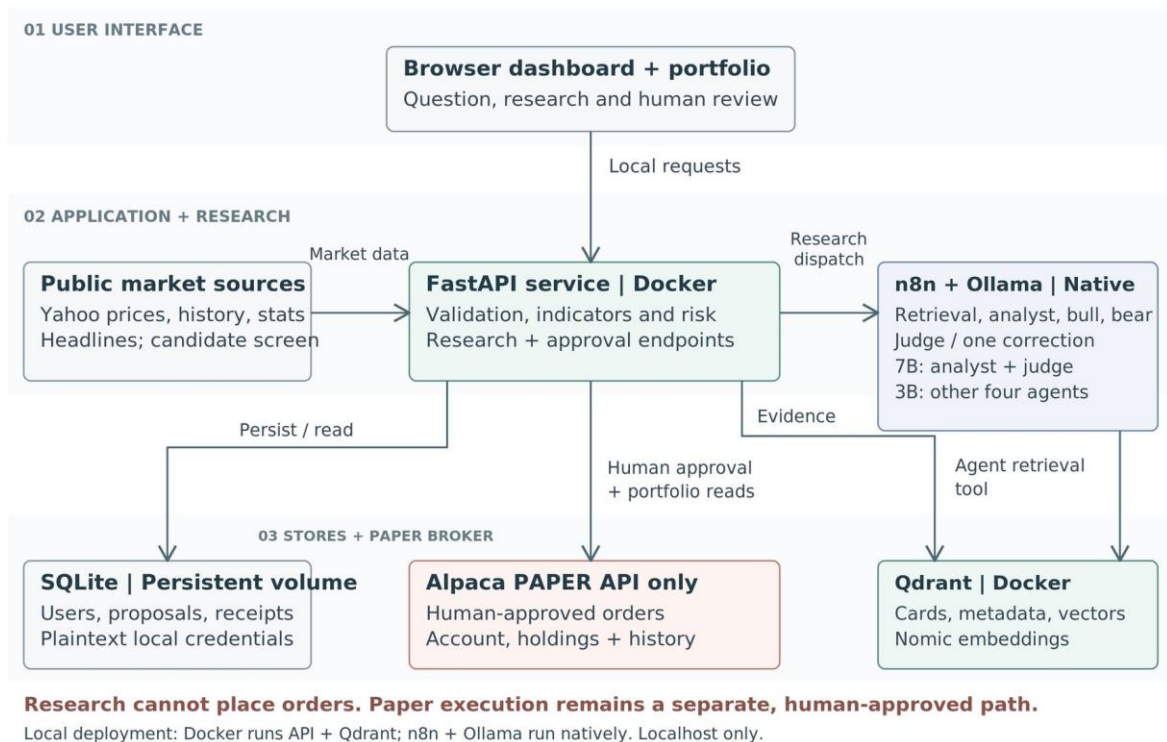


Figure 7.1: System components and the independent paper-execution boundary. Stores support the service and workflow; they are not autonomous trading actors.

7.2 TOOLS

Role-specific prompts combine market snapshots with retrieval instructions. Qdrant is the evidence tool; the models interpret and debate the retrieved context. Python computes indicators, validates market data and checks proposal limits and broker capacity, while the Alpaca paper API is the only broker interface. Models generate text and judgments; they do not validate risk.

7.3 MEMORY

- Evidence memory — dated source cards embedded with Nomic and stored in Qdrant with IDs, dates and ticker metadata.
- Persistent state — SQLite stores users, proposals and receipts in a persistent ledger volume.
- Broker-owned state — equity, positions, orders and history are read from the Alpaca paper account independently.

These are external state stores, not learned conversational memory.

7.4 REASONING FLOW

Separate roles make challenge and critique visible in the execution logs. A single agent would be simpler and faster, but its internal challenge would be harder to inspect. More roles also add latency and can repeat the same errors. Debate should therefore be treated as a design choice to test, not as proof of higher accuracy.

Correction is limited to one pass. Resolved model tiers are returned for audit, while legacy single-model requests support the benchmark. A 3B correction model can alter a 7B answer, so the interactive split needs its own evaluation; the published single-model results do not answer that question.

Current answer-format limitation. The parser accepts Markdown headings but can also accept missing sections with the fallback summary “Analysis Complete”. It overwrites a mismatched separate summary. This differs from earlier strict-validation descriptions and does not establish that every answer is readable or consistent. Clear summary validation and honest error handling remain necessary.

7.5 AGENT INTERACTIONS

The API resolves the question and prepares the evidence context; n8n then sequences retrieval, analysis, bull-bear debate, critique and correction. The research agents never call the broker. Proposal creation and approval are separate user actions, so the research workflow cannot directly place an order, even when its conclusion is wrong.

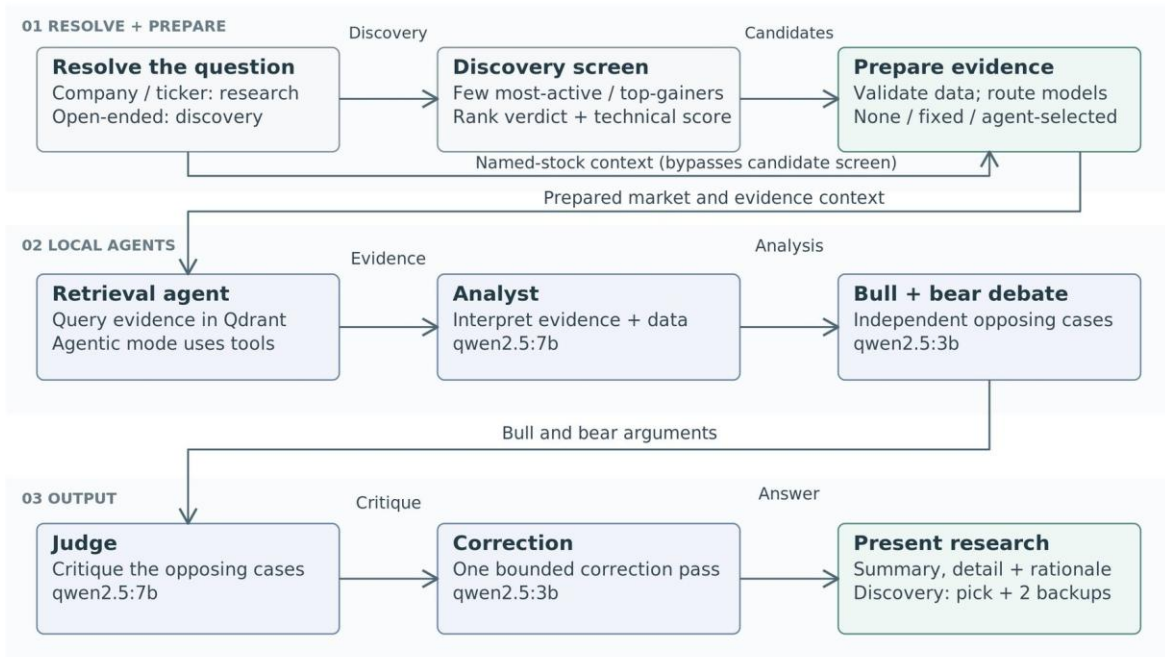
8. AGENT WORKFLOW AND ORCHESTRATION

8.1 WORKFLOW DESIGN

Named-stock research and small-candidate discovery share a bounded research pipeline, shown in Figure 8.1. A named-stock request bypasses the candidate screen; an open-ended request first passes through the discovery screen. Prepared market and evidence context then flows through the retrieval agent, analyst, bull and bear debate, judge and one correction pass before the research is presented.

Research and discovery flow

Named-stock research and small-candidate discovery share a bounded research pipeline.



Research is not an order. Current discovery mapping uses fixed retrieval.

AGENTTRADE | Paper trading only | Source: current report and supplied system diagrams

Figure 8.1: Workflow design — named-stock research and small-candidate discovery.
Discovery uses fixed retrieval under the current browser selection mapping.

8.2 TASK DECOMPOSITION

The API resolves the question: a company or ticker leads to research, while an open-ended question leads to discovery. For discovery, a few most-active and top-gainer candidates are screened, each is researched, and the verdict and technical score rank them into one pick and up to two backups. Evidence preparation validates the data and routes the models for no retrieval, fixed retrieval or agent-selected retrieval.

8.3 AGENT COORDINATION

n8n sequences retrieval, analysis, bull-bear debate, critique and correction. Timed progress pills are a user-interface aid, not evidence that a workflow node has completed. Actual traces establish tool and model execution.

8.4 ORCHESTRATION IMPLEMENTATION

Only one intended research workflow should be published on the agenttrade-analyze webhook. Re-importing creates copies and can leave older logic answering. The research_clean.json variant uses localhost where research.json uses 127.0.0.1, and the files differ. The chosen file and credentials should be checked on the host; both variants should not be published on the same webhook.

9. AGENTIC RAG IMPLEMENTATION

9.1 RETRIEVAL ARCHITECTURE

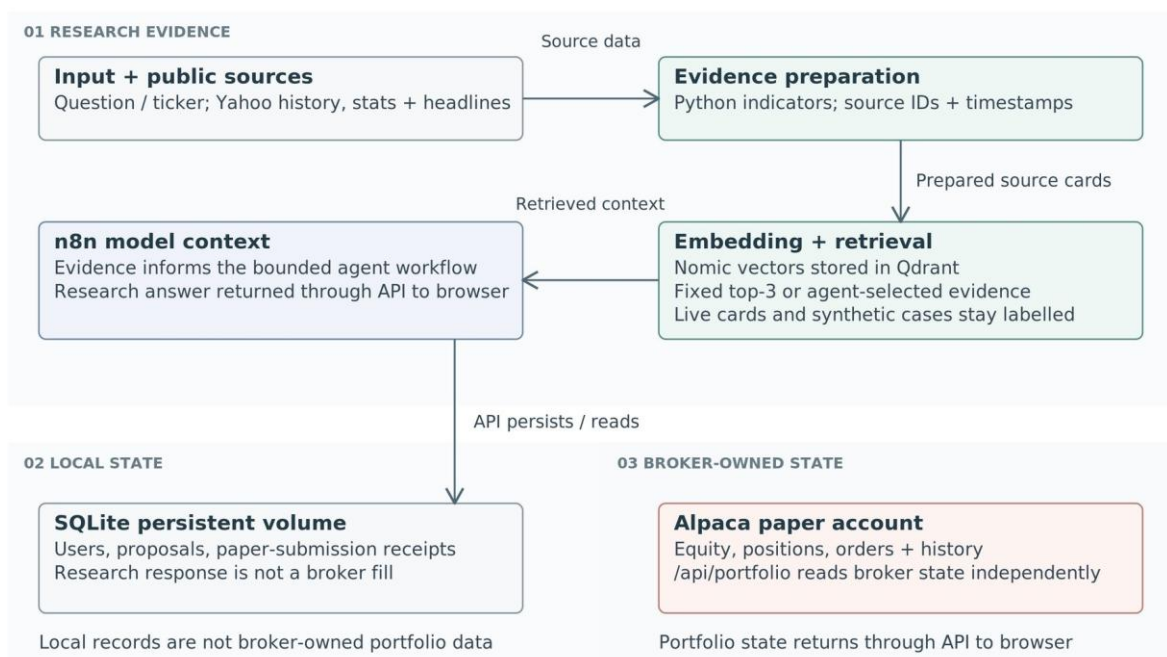
Source cards are embedded with Nomic and stored in Qdrant with IDs, dates and ticker metadata; ticker tags allow company-scoped retrieval. Three retrieval conditions are compared, as summarised in Table 9.1.

Table 9.1: Controlled retrieval comparison

Condition	Retrieval and generation
No RAG / none	Bypass retrieval; no external evidence is supplied.
Basic RAG / fixed	Python retrieves fixed top-3 context as fixed_evidence.
Agentic RAG	The retrieval agent formulates a query and uses Qdrant; query reformulation is possible.

Data and state flow

Evidence context, local proposal state and broker-owned portfolio state have different owners.



Synthetic classroom cases are not financial facts. No conversational learning is claimed.

AGENTTRADE | Paper trading only | Source: current report and supplied system diagrams

Figure 9.1: Evidence preparation, local proposal state and broker-owned state serve different purposes.

9.2 KNOWLEDGE RETRIEVAL

In agentic mode the retrieval agent chooses the evidence search and inspects the metadata, and tool use is expected rather than a written claim that retrieval occurred. Unlike classical fixed RAG, the query can be reformulated. Under the current browser mapping, however, discovery requests use fixed retrieval, so agentic retrieval applies to named-stock research. Live cards and synthetic cases stay labelled, and a synthetic classroom case is not a financial fact.

9.3 GENERATION PROCESS

Generation follows the bounded role sequence. The analyst interprets the retrieved evidence and indicators and prepares an initial view; the bull and bear roles develop independent opposing cases; the critic judge assesses support, conflicting cases and the decision; and one correction pass revises the answer using the critic feedback. The result is returned through the API as a summary, detail and audit fields. Research output is never a broker fill.

9.4 RETRIEVAL-QUALITY EVALUATION

Term accuracy, citation presence and a rationale proxy do not directly measure source relevance or expert financial accuracy. Human annotation and traces are needed before deciding that a retrieval policy is better. In this project the three conditions are compared on the same 25 frozen questions, as described in Section 11.

10. SYSTEM IMPLEMENTATION

10.1 FUNCTIONAL PROTOTYPE SCREENSHOTS

Research desk. The desk contains the question input, ticker shortcuts, retrieval selection, agent explanation and proposal controls. The initial page and its JavaScript loaded without errors in Chrome. A live question still requires running n8n, Ollama and data access, and external market widgets do not demonstrate that research has completed.

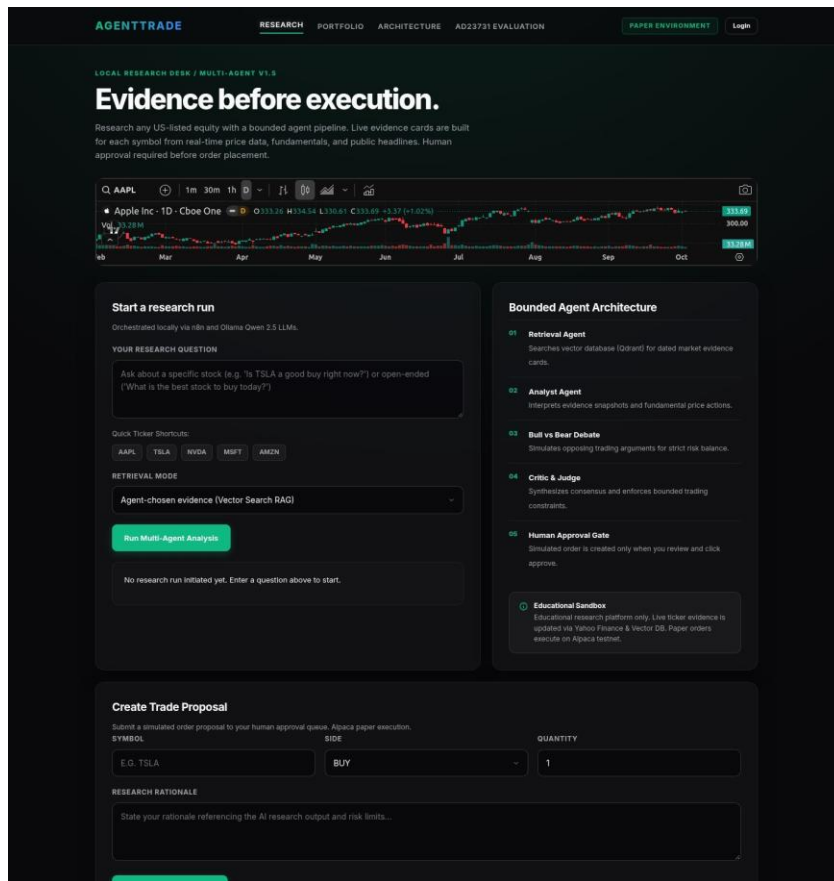


Figure 10.1: Upper portion of the actual dashboard initial state, locally rendered on 4 October 2026. This is an interface screenshot, not a completed model run.

Portfolio and local credentials. The /api/portfolio endpoint reads broker equity, cash, buying power, positions, recent orders and history. A submission receipt is not fill-confirmed performance, and new accounts may have no chart history. The screenshot below therefore establishes the unconfigured interface state, not a return or profit.

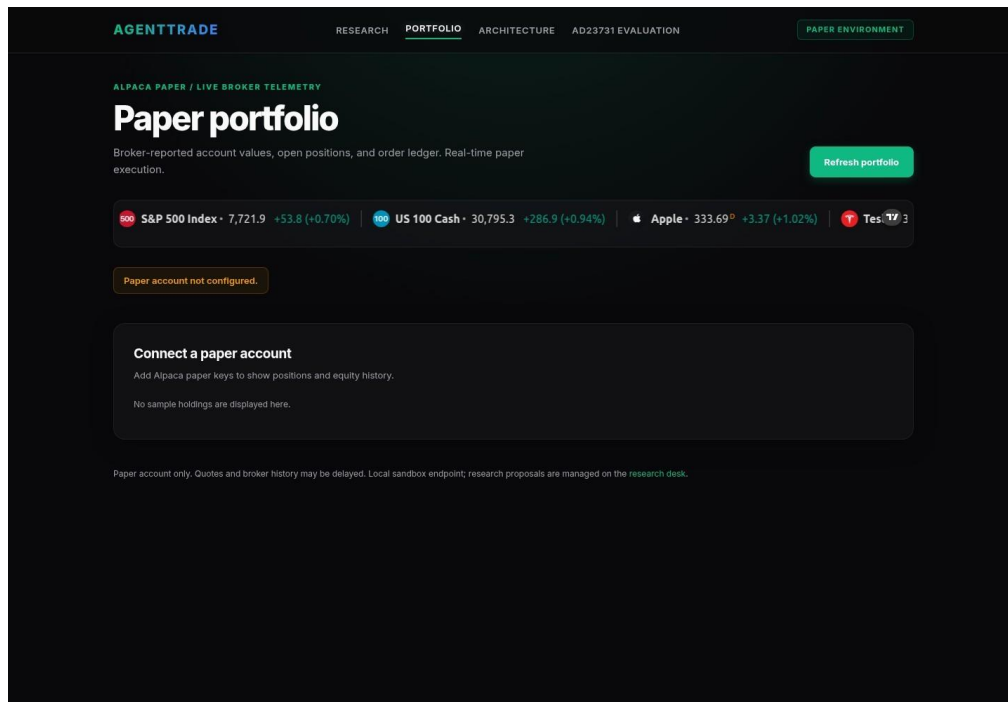


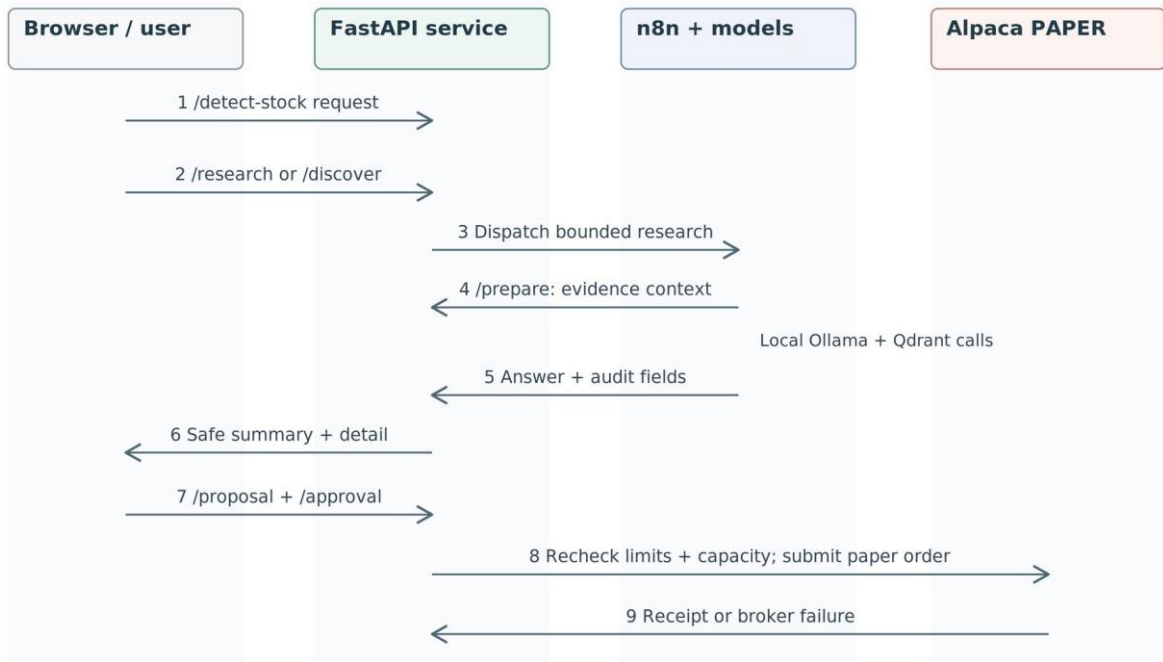
Figure 10.2: Actual paper-portfolio unconfigured state. No sample holdings or investment returns are asserted.

The dashboard includes registration, login and paper-key setup. Passwords and per-user keys are stored in plaintext SQLite. Login does not establish complete endpoint session authorization, and filtering a ledger by username is not an access-control boundary. Keep the host private and localhost-only. Reports, Git and screenshots must not contain actual credentials.

End-to-end request sequence. The research request returns a summary, detail and audit fields through the API. A separate user choice creates a proposal and reaches approval. Portfolio reads obtain broker state independently of research output.

End-to-end request sequence

Research and paper approval are separate request paths. Portfolio reads are independent.



Portfolio reads broker state independently. This is a source sequence, not a verified live order.

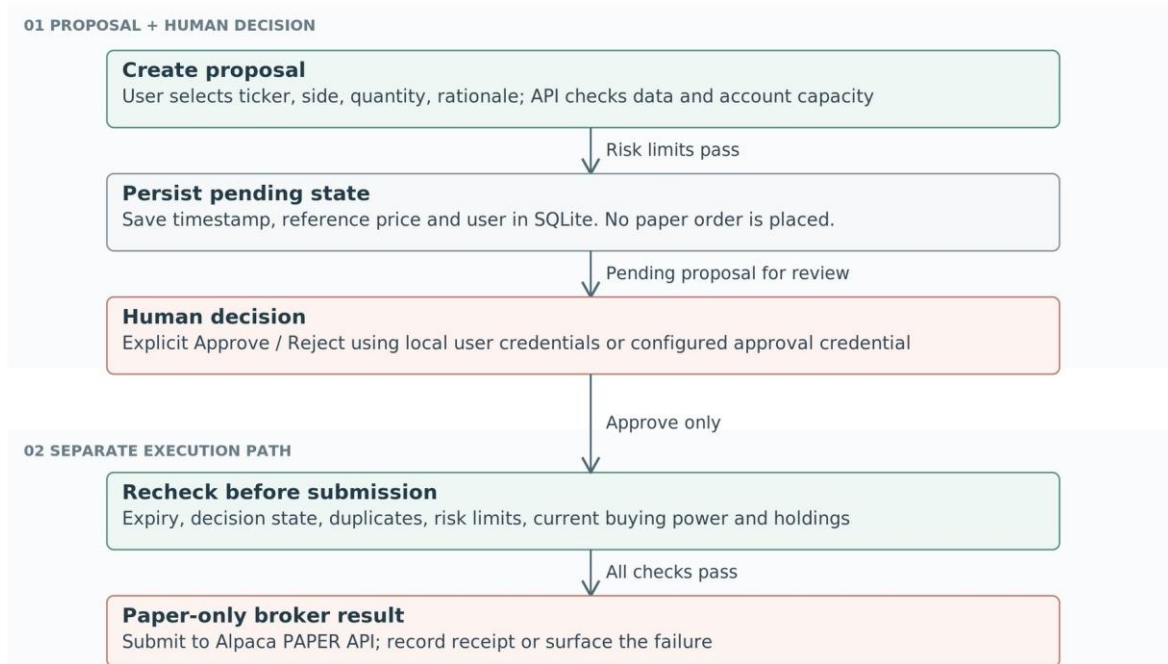
AGENTTRADE | Paper trading only | Source: current report and supplied system diagrams

Figure 10.3: Browser, API, workflow and broker request sequence. This describes source behavior, not a verified live broker submission.

Human review and independent risk controls. The user should inspect the ticker, side, quantity, rationale and account capacity before choosing approval. Model confidence is not permission. Proposal creation requires configured paper keys and capacity checks; the earlier no-key local proposal behavior is not supported. Limits include up to five shares, USD 1,000 notional, no short selling, 30-minute expiry and duplicate checks. Approval repeats the relevant checks, including current buying power and holdings, and surfaces broker failures instead of reporting a false success.

Human approval and paper-order gate

30-minute expiry | Maximum 5 shares and USD 1,000 notional | No short selling



Reject, expiry or failed checks do not place an order. Research has no order path.

AGENTTRADE | Paper trading only | Source: current report and supplied system diagrams

Figure 10.4: Independent human-approval and paper-order path, including expiry, duplicate and account-capacity checks.

Alpaca paper trading is used to simulate orders [6]. No evidence in this report establishes a real broker submission, a filled order or profitable trading. The paper-only boundary must remain in place during local validation.

Rendering patch. The implementation escapes model and news HTML, permits inert formatting and removes active attributes. A harmless local image-event regression test no longer executed after that patch. This was a narrow rendering fix, not a full security audit.

10.2 SOURCE-CODE REPOSITORY

The source code, Docker Compose configuration, n8n workflow JSON files (research.json and research_clean.json) and evaluation scripts are kept together with the project. Repository link: **[to be added by the project team]**. Credentials must not be committed to the repository.

11. EXPERIMENTAL EVALUATION AND RESULTS

11.1 EVALUATION METRICS

Table 11.1: Evaluation metrics

Metric	Meaning and caution
Label-term accuracy	Expected-term matching; a wrong answer may still contain the correct term.
Reasoning proxy	Citation/rationale heuristic, not expert grading.
Citation rate	Detected citation syntax, not verified source support.
Exact consistency	Normalized answer equality; paraphrases may score zero.
Success / errors	Request completion counts; HTTP success is not answer correctness.
Latency	Timing is recorded by the evaluator, but no raw verified timing table was supplied.

11.2 TESTING METHODOLOGY

The experiment uses 25 fixed questions, two single-model configurations, three retrieval modes and three repeats: $25 \times 2 \times 3 \times 3 = 450$ workflow requests. Each request includes several model calls. Qwen2.5 3B is compared with Llama3.2 3B; this is not an evaluation of the interactive 7B/3B split.

The same frozen cards, model versions, prompts and embeddings should be used, starting from a clean collection, with answers preserved and retrieval traces inspected. Source support should be annotated manually. A one-repeat pipeline smoke run cannot establish response consistency.

11.3 PERFORMANCE RESULTS

The project report records a run start of 3 October 2026 at 16:24:59 UTC and six rows totaling 450 successful requests. The raw CSV, summary JSON and traces were not supplied. The values in Table 11.2 are reported, not independently rerun or verified, and no model benchmark was run for this restyled report.

Table 11.2: Reported results, with scores on a 0–1 scale

Model	Mode	OK / total	Accuracy	Reason.	Citation	Consist.
Qwen 3B	none	75 / 75	0.307	0.000	0.013	0.000
Qwen 3B	fixed	75 / 75	0.707	0.000	0.000	0.000
Qwen 3B	agentic	75 / 75	0.640	0.013	0.053	0.000
Llama 3B	none	75 / 75	0.293	0.000	0.000	0.000
Llama 3B	fixed	75 / 75	0.893	0.000	0.000	0.000
Llama 3B	agentic	75 / 75	0.747	0.027	0.027	0.000

Models: qwen2.5:3b and llama3.2:3b. Reported errors are zero per row. No latency or profit values were supplied. Values are preserved from the source report reviewed on 4 October 2026.

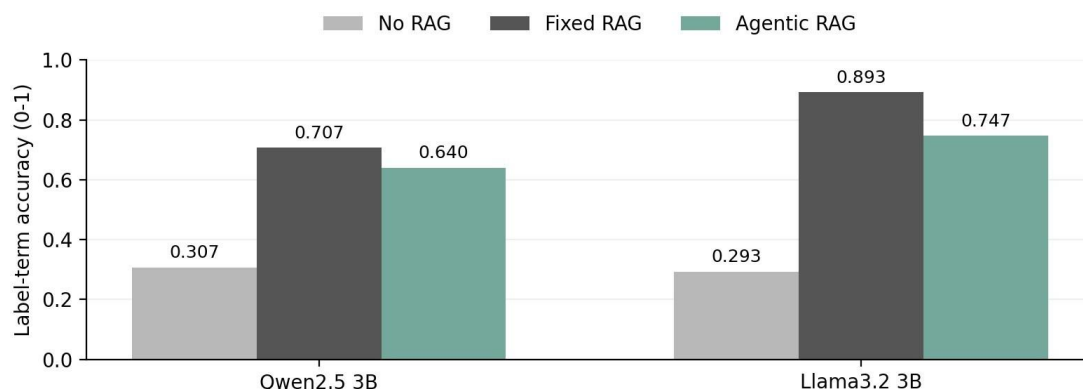


Figure 11.1: Label-term accuracy calculated from Table 11.2. The graph is not a backtest or a measure of financial return.

Fixed RAG has higher reported term accuracy than agentic RAG for both models: 0.707 versus 0.640 for Qwen and 0.893 versus 0.747 for Llama. No-RAG is lower. The table does not support a general claim that agentic retrieval outperforms fixed retrieval.

Citation rates and reasoning proxies are low, and exact consistency is 0.000 in every row. Formatting, metric design or actual instability may contribute; raw answers and traces are needed to distinguish them. The small synthetic task set is not investment evidence.

Table 11.3 lists the implementation checks reported on 4 October 2026.

Table 11.3: Observed implementation checks

Check	Reported outcome
Dependencies / Python	Pinned API dependencies installed; source compiled.
Scripts / initial pages	Scripts parsed; research, portfolio and evaluation loaded without JS errors.
Initial API endpoints	/, /portfolio, /architecture, /evaluation, /health and /api/portfolio returned 200 locally.
Diagnostic offline checks	68 passed and 17 failed after a diagnostic-only continuation; not a clean pass.
Official smoke_test.py	Fails and crashes on an old exact-string UI assertion; tests need revision.
Injection regression	A harmless model/news test flag executed before the rendering patch, but not afterward.
Docker / models / broker	No complete runtime verification: no Docker daemon, running n8n/Ollama or actual paper credentials.

Several failures are stale after changes to discovery, the interface and key-required proposals. Others identify summary-validation drift. The suite must not be called green, and fake test data must not be presented as a live research run. Workflow variants also need reconciliation.

11.4 SUCCESS-METRIC ACHIEVEMENT

Table 11.4: Success criteria

Success criterion	Expected outcome	Achieved
Retrieval benefit	Retrieval raises label-term accuracy over no-RAG	Yes (reported: 0.707 vs 0.307 Qwen; 0.893 vs 0.293 Llama)
Agentic advantage	Agentic RAG scores above fixed RAG	No (0.640 vs 0.707 Qwen; 0.747 vs 0.893 Llama)
Execution safety boundary	Research cannot place orders; approval is separate	Implemented by design; no live broker verification
Citation reliability	Claims linked to verified sources	Not established; citation rates are detected syntax only
Test suite	Clean pass of the smoke and offline tests	No (68 passed, 17 failed in a diagnostic run)
End-to-end runtime	Docker, n8n, Ollama and paper broker verified	Not verified
Efficiency	Latency and cost within acceptable limits	Not measured; no latency values supplied

12. COMPARISON, INNOVATION AND SCALABILITY

12.1 COMPARISON WITH BASELINE

No-RAG isolates model-only answers. Fixed RAG tests deterministic top-3 context, while agentic RAG tests query choice and tool use. Quality and cost should be compared empirically; the current table favors fixed retrieval on strict label-term accuracy. The quantitative comparison is presented in Section 11.

12.2 ADVANTAGES OF THE PROPOSED SYSTEM

- Local inference reduces remote-model disclosure, although data and broker calls remain external.
- Separating research from approval reduces accidental execution, but identity and secret handling still need work.
- Bounded correction is inspectable, while multiple model calls increase latency.
- A readable interface helps review; timed progress cannot replace authoritative execution traces.

12.3 INNOVATION

The contribution is the combination of the following design elements. This is a project design contribution, not a claim that the idea is unprecedented.

- Inspectable role separation — retrieval, analyst, bull, bear, critic and one bounded correction.

- Live ticker evidence alongside a frozen synthetic benchmark corpus.
- Deterministic risk checks independent of the language models.
- Human approval as the only route to a paper order.
- Discovery that reduces a broad question to a small candidate screen.
- Local evaluation of no-RAG, fixed-RAG and agentic-RAG conditions.

12.4 SCALABILITY

The prototype targets one machine. Wider use needs queues, request isolation, retrieval monitoring, per-user authorization, protected secrets, rate limits, data rights and a stronger database design. More discovery candidates multiply inference and data cost. No throughput or capacity claim is made.

12.5 REAL-WORLD APPLICABILITY

Financial research use would also require broader sources, filings or full-article verification and expert review. Paper integration demonstrates an execution boundary; it does not authorize real investment execution. Within a classroom setting, the system lets students compare retrieval policies and lets evaluators inspect a human-in-the-loop trading simulation.

13. CONCLUSION AND FUTURE WORK

13.1 CONCLUSION

AGENTTRADE implements bounded research, evidence retrieval, candidate discovery and human-approved paper execution. Initial UI and API state render correctly, but the recorded test failures and missing benchmark artifacts prevent a stronger end-to-end claim. The reported comparison is useful because it also shows where the agentic design did not beat the simpler fixed-retrieval baseline.

13.2 LIMITATIONS

- The corpus and question set are small and synthetic, and mixed live cards can contaminate evaluation.
- Expected terms and citation syntax can distort scoring.
- The results cover model and RAG comparisons, but not expert reasoning accuracy, readability, effective tool use, the two-tier 7B/3B split or trading usefulness.
- Plaintext credentials, incomplete authorization and locally exposed approval credentials prevent safe public multi-user deployment.
- The rendering patch fixes one class of unsafe output but does not remove these limitations.
- Raw answers, summary.json and traces were not supplied, so the benchmark values are not independently verified.

13.3 FUTURE WORK

- Update behavioral tests and restore clear summary validation without concealing model errors.
- Preserve raw answers, summary.json, model versions and traces; add human source-support scoring.
- Evaluate 7B/3B routing and discovery separately, including measured latency and quality.
- Hash passwords, protect keys and enforce session and ownership checks before network expansion.
- Reconcile workflow variants and replace timed progress with actual execution status.

Local validation checklist. Use the intended implementation version, configure secrets locally without including them in source files, run docker compose config, then rebuild the API. Install the native models and publish one intended research workflow. Run the smoke suite and record every failure. Check a named stock, open-ended discovery, proposal rejection and expiry, and paper approval. These are required checks, not claimed passes.

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